EXPLORATORY DATA ANALYSIS PROJECT

Employee Attrition & Workforce Analytics

Key drivers of voluntary turnover and retention strategies using Python

IBM HR Dataset (1,470 Records) Pandas & Seaborn Submission

WORKFORCE BASELINE

Overall Company Attrition Benchmark

TOTAL EMPLOYEES

1,470

Complete surveyed workforce records.

TURNOVER RATE

16.1%

237 voluntary employee resignations.

RETAINED STAFF

83.9%

1,233 stable retained employees.

Baseline metric established for comparative segment analysis.

PRIMARY OPERATIONAL RISK

Overtime Multiplies Resignation Risk by 3x

WITH OVERTIME

30.5%

Severe burnout and fatigue.

WITHOUT OVERTIME

10.4%

High retention under standard hours.

Overtime employees represent 127 departures out of only 416 staff members.

FINANCIAL DRIVERS

Low Salaries Drive Early Resignations

MONTHLY SALARY GAP

$2,002

Median income difference per month.

Job Levels 1 & 2 Vulnerability

Most exits occur below $5,000 monthly pay.

Senior Leadership Stability

Job Levels 4 and 5 experience near-zero attrition.

Retained staff earn $5,204 median salary versus $3,202 for departing staff.

DEPARTMENT & ROLE ANALYSIS

Frontline Roles Face Highest Turnover



Sales Representatives experience nearly 4 departures out of every 10 staff members.

WORKPLACE ENVIRONMENT

Commute Distance & Low Morale

WORK-LIFE BALANCE RATING = 1

31.3%

Doubles the resignation rate.

AVERAGE COMMUTE DISTANCE

10.6 Miles

Departing staff live further away.



Long travel times combined with heavy workloads accelerate employee fatigue.

HR ACTION PLAN

Three Data-Driven Retention Initiatives

Cap Overtime Hours

Enforce weekly overtime limits and offer paid compensatory time off.

Benchmark Entry Pay

Increase Job Level 1 and 2 salaries to market median rates.

Hybrid Work Policy

Provide 2 to 3 remote days for employees commuting over 10 miles.

Target outcome: Reduce baseline voluntary attrition toward 10% within four quarters.

Image Sources



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